5.1 Introduction to Probability

1. Probability of an Events

$$P(A) = \frac{\text{Number of favorable outcomes (Outcomes in A)}}{\text{Total number of possible outcomes}}$$

*Note: This counting formula works when all outcomes are equally likely

Range of Value: $0 \leq P(A) \leq 1$

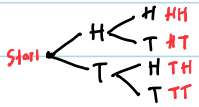
P(A)	Meaning
0	Impossible, it never happens
1	Certain, it always happen (100%)
0.3	30% Chance A happen

2. Sample Space and tree diagram

The sample space S is the set of all possible outcomes.

One coin: $S = \{H, T\}$

Two Coin: $S = \{HH, HT, TH, TT\}$



05.2 Probability Formulas, Symbols & Notations – Marginal, Joint, and Conditional Probabilities

1. Marginal probability – a single event

The probability of one event, regardless of any others:

$$P(A) = \frac{\text{Favorable Outcomes in event A}}{\text{Total Outcomes in Sample Space}}$$

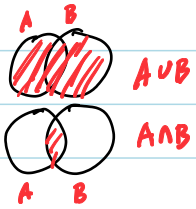
example 1: • Sample Space $S = \{1,2,3,4,5,6,7,8,9\}$

• "Event" $A = \{1,2,3,4\}$ $B = \{6,7,8\}$ $C = \{1,7,8,9\}$

$P(A) = \frac{4}{9} = 44.44\%$ $P(B) = \frac{3}{9} = 33.33\%$. $P(C) = \frac{4}{9} = 44.4\%$

2. Union and Intersection Symbols

Symbols	Name	Meaning
$A \cup B$	Union	Everything in A or B (or both), listed once
$A \cap B$	Intersection	Only what is in both A and B



Example 1:

$A \cup B = \{1,2,3,4,5,6,7,8\}$

$A \cap B = \{\text{None}\}$

$B \cup C = \{1,6,7,8,9\}$

$B \cap C = \{7,8\}$

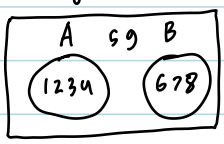
3. Union Probability – "A or B" Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(A \cup B)$: $P(A$ or $B)$ — $P(A \cap B)$: $P(A$ and $B)$

The Subtraction stops $A \cap B$ being counted twice.

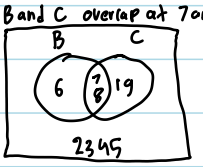
• Mutually exclusive events can't happen together, so $A \cap B \neq \emptyset$ and $P(A \cap B) = 0$: **Mutually Exclusive A, B**



$P(A \cup B) = \frac{7}{9} = 0.778 = P(A) + P(B) = \frac{4}{9} + \frac{3}{9}$

• Example overlap at 7 and 8: $B \cap C = \{7,8\}$ so $P(B \cap C) = \frac{2}{9}$: **Not Mutually Exclusive B, C**

B and C overlap at 7 and 8



$P(B \cup C) = \frac{5}{9} = P(B) + P(C) - P(B \cap C) = \frac{4}{9} + \frac{3}{9} - \frac{2}{9} = \frac{5}{9}$

4. Joint Probability - $P(A \cap B)$ (multiplication rule)

- $P(A \cap B)$ is read "The probability of A and B" (both happen):
  - ↳ Words: $P(A \text{ and } B) = P(A \text{ given } B)\ P(B) = P(B \text{ given } A)\ P(A)$
  - ↳ Symbols: $P(A \cap B) = P(A|B)\,P(B) = P(B|A)\,P(A)$

On the **Independent events** (one doesn't affect the others): $P(A|B) = P(A)$ and $P(B|A) = P(B)$, so

Words: $P(A \text{ and } B) = P(A)\,P(B)$

Symbols: $P(A \cap B) = P(A)\,P(B)$

- Independent or dependent?
  - Situation: Draw a red marble, **Put it back**, then draw blue one
    - Type: Independent.
    - Why: Replacing the marble restore the bag.
  - Situation: Draw a red marble, **keep it**, then draw blue one
    - Type: Dependent.
    - Why: The bag now has one fewer marble, so the second probability change.
  - Situation: Toss a coin Twice
    - Type: Independent
    - Why: The first toss doesn't affect a second toss.

- Notation Warning: $P(A \cap B)$ vs $P(AB)$
  - $P(A \cap B) = P(B \cap A)$: both happen, no order implied.
  - Some books write $P(AB)$ for "A first, then B" and $P(BA)$ for "B first, then A". These are not always equal.
  - Read the context: are the events simultaneous, or does one come before the other?

5. Conditional Probability - "A given B"

$P(A|B)$ is the probability of A **given that** B **has already happened**.

we know that $P(A \cap B) = P(A|B)\,P(B) = P(B|A)\,P(A)$

So

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Conditional = Joint / Marginal

6. Bayes' Theorem

Replacing $P(A \cap B)$ with $P(B|A)\,P(A)$

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Replacing $P(A \cap B)$ with $P(A|B)\,P(B)$

$$P(B|A) = \frac{P(A|B)\,P(B)}{P(A)}$$

7. Negation (Complement) Probability

Words: $P(\text{not } A) = 1 - P(A)$

Symbols: $P(A') = 1 - P(A)$

$P(A')$ is read "The probability of Not A". The complement is also written A^c (The most common form in textbook), $\bar{A}$ or $\sim A$

example 1:

$P(A') = 1 - P(A) = 1 - \frac{4}{9} = \frac{5}{9}$

$P(B) = 0.35$ then $P(B') = 1 - 0.35 = 0.65$, a 65% chance that B doesnot occur.

>Formula Summary

Type	Words	Symbols
Marginal	$P(A) = \frac{\text{favorable}}{\text{total}}$	(same)
Union (or)	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Union, mutually exclusive	$P(A \text{ or } B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B)$
Joint (and)	$P(A \text{ and } B) = P(A \text{ given } B)P(B) = P(B \text{ given } A)P(A)$	$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$
Joint, independent	$P(A \text{ and } B) = P(A)P(B)$	$P(A \cap B) = P(A)P(B)$
Conditional	$P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}$	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
Bayes	$P(A \text{ given } B) = \frac{P(B \text{ given } A)P(A)}{P(B)}$	$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$
Complement	$P(\text{not } A) = 1 - P(A)$	$P(A') = 1 - P(A)$

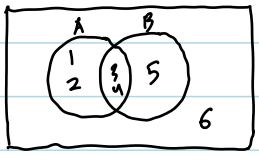
05.3 Probability of Complementary Events

Setup - rolling one six-sided die

- Sample Space $S = \{1,2,3,4,5,6\}$
- A = number at most 4 = $\{1,2,3,4\}$
- $B = \{3,4,5\}$

1. P(A and B) - Intersection - $P(A \cap B)$

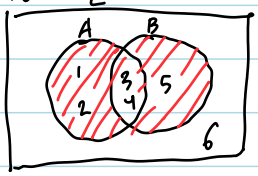
$A \cap B = \{3,4\}$ (the outcomes in both sets)



$$P(A \cap B) = \frac{2}{6} = \frac{1}{3} = 0.33$$

2. P(A or B) - Union - $P(A \cup B)$

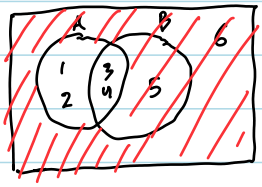
$A \cup B = \{1,2,3,4,5\}$ (the outcomes in either set; only 6 is missing)



$$P(A \cup B) = \frac{5}{6}$$

3. P(not A) - Complement - $P(A')$

The complement A' is everything in S that is not in A: $A' = \{5,6\}$



4. P(A) and the complement rule

$$P(A) = \frac{4}{6} = \frac{2}{3}$$

$$P(A) + P(A') = \frac{2}{3} + \frac{1}{3} = 1$$

↓ event — ↓ complement of event A

An event and its complement together cover the whole sample space, so their probabilities add to 1:

$$P(A') = 1 - P(A)$$

05.4 Conditional Probability with Venn Diagrams and Contigency Tables

- **What it Means**

$P(A|B)$ is the probability that A happens given that B has already happen

Formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ Word: $P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}$

Shortcut: ask "how much of B is also A?" Once B is known to have happened,

$P(A|B) = \frac{\text{Count in both A and B}}{\text{Count in B}}$ → B becomes the new sample space.

- **Example 1 - a die**

$S = \{1,2,3,4,5,6\}$, $A = \text{odd} = \{1,3,5\}$, $B = \{3,4,5\}$

Shortcut: of the 3 outcomes in B, two (3 and 5) are in A:

$$P(A|B) = \frac{2}{3}$$

Formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3} \checkmark$$

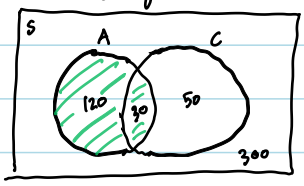
- **Example 2 - Venn diagram (algebra and Chemistry)**

A school has 500 students, 150 take algebra (A), 80 take chemistry (C), and 30 take both.

The four region of the venn diagram (two overlapping circles inside a box of 500):

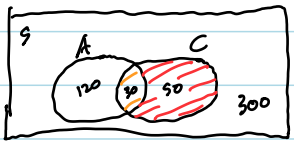
Region	Count
Algebra only (A but not C)	150 − 30 = 120
Both ($A \cap C$, the overlap)	30
Chemistry only (C but not A)	80 − 30 = 50
Neither (outside both circle)	500 − 200 = 300
Total	500

(a) $P(\text{taking Algebra}) = P(A)$



$P(A) = \frac{150}{500} = \frac{3}{10} = 0.3$

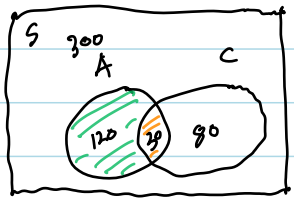
(b) $P(\text{Algebra} \mid \text{Chemistry}) = P(A|C)$



$P(A|C) = \frac{30}{80} = 0.35$

$= \frac{P(A \cap C)}{P(C)} = \frac{30/500}{80/500}$

(c) $P(\text{Chemistry} \mid \text{Algebra}) = P(C|A)$



$P(C|A) = \frac{P(C \cap A)}{P(A)}$

$= \frac{30/500}{150/500} = 0.2$

• Example 3 - Contingency Table (birds in a Zoo)

200 birds: - 70 male with brown eyes $|M \cap Br| = 70$ $n(M \cap Br)$

- 100 femal with brown eyes $|F \cap Br| = 100$ $n(F \cap Br)$
- 20 male with brown eyes $|M \cap Br| = 20$ $n(M \cap Br)$
- 10 female with blue eyes $|F \cap Bl| = 10$ $n(F \cap Br)$

* $|\cdot|$ = Cardinality (number of elements in a set)
* $n(\cdot)$ = Cardinality Function

> Birds

Eye color / Gender	Br Brown Eyes	Bl Blue Eyes	Total
Male	70	20	90
Female	100	10	110
Total	170	30	200

a) P(female)

$$P(F) = \frac{110}{200}$$

b) P(male and brown eyes)

$$P(M \cap Br) = \frac{20}{200}$$

c) P(female | brown eyes)

$$P(F|Br) = \frac{100}{170} \quad \text{or} \quad P(F|Br) = \frac{P(F \cap Br)}{P(Br)} = \frac{100/200}{170/200}$$

d) P(blue eyes | Female)

$$P(Bl|F) = \frac{10}{110}$$

e) P(male | Blue eyes)

$$P(M|Bl) = \frac{P(M \cap Bl)}{P(Bl)} = \frac{20/200}{30/200} = 0.67$$

Joint vs conditional probability in a contingency table

Orange = numerator. Blue outline = denominator. Light blue = the row or column you restrict to.

(b) P(male and brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

70 / 200 = 0.35

joint: divide by the grand total

(c) P(female | brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

100 / 170 ≈ 0.588

given brown: divide by the column total

(e) P(blue | female)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

10 / 110 ≈ 0.091

given female: divide by the row total

The numerator is always one cell. What changes is the denominator: the grand total for a joint probability, or one row or column total for a conditional probability.

Key takeaways

- $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, or with counts, $\dfrac{n(A \cap B)}{n(B)}$.
- The "given" event becomes the **denominator**, the new reduced sample space.
- In a contingency table, a conditional probability uses **one row or column total** as the denominator. A joint probability uses the **grand total**.
- $P(A \mid B)$ and $P(B \mid A)$ are usually different.